Large Language Models as Nonlinear Dynamical Systems: The Three Clocks of Depth, Training, and Generation

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Abstract

A transformer language model is iterated in three ways. Over layers, the residual stream of each token is updated step by step. Over optimizer steps, the weights evolve under gradient descent. Over generated tokens, the model feeds its own samples back as input. We call these the three clocks of a language model: layer time, training time and generation time. Each clock has its own literature on clustering, edge of stability, loss spikes, grokking, repetition loops and temperature transitions, and these literatures rarely cite each other. We place the three clocks in one framework of coupled maps, review about 180 papers across them, and condense the results into the Three-Clock Dictionary, which maps fifteen concepts from nonlinear dynamics onto observed behavior on each clock and marks whether each correspondence is proved, measured or conjectured. Three CPU-scale experiments on open models illustrate the dictionary. In public Pythia checkpoints, training makes every layer stretch perturbations more strongly, while the forward pass as a whole changes little because the expanding directions of consecutive layers become less aligned. Gradient descent on a small transformer stays at the edge of stability across two decades of step size, passing from a period-2 orbit to irregular oscillation before it diverges. In GPT-2 and Pythia-160M, escape from repetition loops follows an Arrhenius law in temperature with the logit gap as the barrier (correlation 0.997 between fitted and measured barriers), and the barrier grows from cycle to cycle in 71 of 73 loops, so loops become harder to leave the longer they last. We close with twelve open problems and release code, data and a maintained reading list.

1 Introduction

A trained transformer language model is usually described as a function that takes a context and returns a distribution over the next token. For most engineering purposes this description is enough.[1] It hides, however, that the model is used as an iterated map in at least three different ways, and each of these iterations has its own literature, its own vocabulary and its own catalogue of strange behavior.

The first iteration runs over depth. Each residual block adds an update to the hidden state of every token, so a stack of L blocks is a discrete-time dynamical system with the layer index as time (E, 2017; Haber & Ruthotto, 2017; Lu et al., 2017; Chen et al., 2018), and attention couples the tokens into a system of interacting particles (Lu et al., 2019; Sander et al., 2021b; Geshkovski et al., 2023b). The second iteration runs over optimization steps. Gradient descent is a map on parameter space, and at the step sizes used in practice it sits at the boundary of linear stability (Cohen et al., 2021), produces loss spikes (Chowdhery et al., 2022; Wortsman et al., 2023) and goes through abrupt transitions such as grokking (Power et al., 2022) and the formation of induction heads (Olsson et al., 2022). The third iteration runs over generated tokens. An autoregressive model feeds its own samples back as input, so decoding is a stochastic dynamical system on the space of contexts, and its best known pathology, the repetition loop (Holtzman et al., 2019; Xu et al., 2022), has recently been analyzed as an attractor (Wang et al., 2025; Jones, 2026).

[1] Use of AI tools: an AI assistant (Claude, Anthropic) helped search and verify the literature, draft and edit text, and write and run the experiment code, under the author's direction. All references were checked against arXiv or Crossref records and all reported numbers come from the released code. The author takes full responsibility for the content.

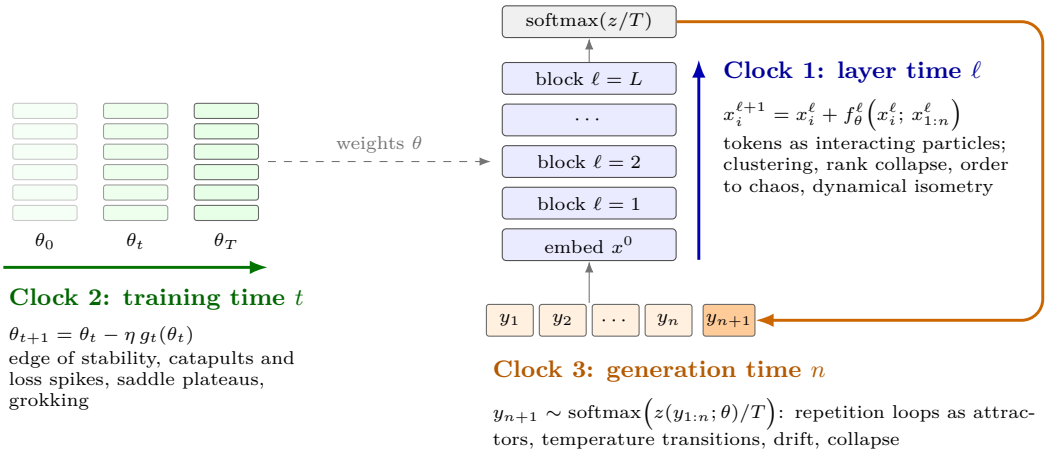


Figure 1: The three clocks of a transformer language model. **Clock 1** (blue) iterates the residual update over layers for a fixed input and fixed weights. **Clock 2** (green) iterates the optimizer over the weights; it hands its current weights to the model. **Clock 3** (orange) iterates sampling: each new token is appended to the context and the whole stack runs again. One tick of clock 3 contains a full run of clock 1, and clock 2 changes the map that both of them iterate. Typical behavior studied on each clock is listed next to it; Table 1 maps it onto dynamical-systems concepts.

We call these three iterations the *three clocks* of a language model, with layer time ℓ, training time t and generation time n (Figure 1). The clocks are coupled, since the parameters produced by clock 2 define the map iterated by clock 1, and every tick of clock 3 calls a full run of clock 1. They also run at very different rates, since the weights do not move during generation and each weight update in training averages over many forward passes. Separated time scales of this kind are the standard setting of slow-fast dynamical systems (Strogatz, 2018), and they explain why three literatures could grow up around one object with few citations between them.

In the current work we bring these literatures together, with the following contributions.

1. **A unified framework.** We formalize the three clocks as coupled maps (Section 2) and review the dynamical-systems results for each clock (Sections 3, 4 and 5), drawing on about 180 papers from machine learning theory, interpretability, optimization, statistical physics and applied mathematics.
2. **The Three-Clock Dictionary.** Table 1 maps fifteen concepts from nonlinear dynamics (fixed points, limit cycles, period doubling, Lyapunov exponents, basins, metastability, noise-driven escape, synchronization, Koopman linearization and others) onto observed behavior on each clock, with the evidence level of every entry. An empty cell marks a concept that no study we found has tested on that clock, and each such cell defines a research question.
3. **Three small experiments** on open models that run on a laptop CPU (Section 7). We measure how the per-layer Lyapunov profile of the residual stream changes over training in the public Pythia checkpoints (Biderman et al., 2023), we compute a learning-rate bifurcation diagram of a small transformer trained by full-batch gradient descent, and we show that escape from repetition loops in GPT-2 and Pythia follows an Arrhenius law with the logit gap as the Kramers barrier (Kramers, 1940), a barrier that grows while the model stays in the loop.
4. **Twelve open problems** (Section 8), each stated with a first test.

Several papers develop one of these views in depth, and we rely on them throughout. Geshkovski et al. (2023b) and Castin et al. (2025) develop the mathematics of attention as an interacting particle system over depth. Cohen et al. (2021) and the work that followed analyze gradient descent at the edge of stability. Krishnamurthy et al. (2026) give a stochastic dynamical-systems account of autoregressive generation and of

networks of interacting language models, and Jones (2026) studies chaos and bifurcations in a deterministic feedback version of generation. To our knowledge, no earlier work treats the three clocks side by side in one vocabulary, with an explicit record of which correspondences are proved, which are measured and which are conjectures.

We use the language of dynamical systems in its technical sense wherever a result supports it, and we flag analogies as analogies. For example, we call a loss spike a bifurcation only when a parameter can be shown to cross a stability threshold. The paper is written for machine learning researchers who want a compact entry into dynamical-systems tools, and for dynamicists who want to know which of their tools are already in use on language models. Section 2 is a primer for the first group, and the dictionary in Section 6 is intended for both.

2 The three clocks, and a primer on the tools

2.1 Three coupled maps

Fix a decoder-only transformer (Vaswani et al., 2017) with L residual blocks, model width d and vocabulary $\mathcal{V}$. For a context of n tokens, write $x^\ell = (x_1^\ell, \dots, x_n^\ell) \in \mathbb{R}^{n \times d}$ for the residual stream after block ℓ, and θ for all parameters. The model defines three maps.

Definition 1 (Clock 1: layer time)**.** For fixed θ and fixed input tokens,

$$x^{\ell+1} = x^\ell + f_\theta^\ell(x^\ell), \qquad \ell = 0, \dots, L-1, \tag{1}$$

where f_θ^ℓ contains attention, which couples token i to tokens $j \le i$, and a token-wise MLP. The logits are $z = W_U \,\mathrm{LN}(x_n^L)$.

Definition 2 (Clock 2: training time)**.** For a loss $\mathcal{L}$ and data stream ξ_t,

$$\theta_{t+1} = \theta_t - \eta_t \, P_t \, \nabla_\theta \ell(\theta_t; \xi_t), \qquad t = 0, 1, \dots \tag{2}$$

where P_t is a preconditioner (identity for gradient descent, a diagonal matrix built from gradient moments for Adam) and $\ell(\cdot; \xi_t)$ is the minibatch loss. With full batches the map is deterministic; with minibatches it is a random dynamical system.

Definition 3 (Clock 3: generation time)**.** For fixed θ, a sampling temperature $T > 0$ and a context $s_n = (y_1, \dots, y_n)$,

$$y_{n+1} \sim \pi_T(\cdot \mid s_n) = \mathrm{softmax}\big(z(s_n; \theta)/T\big), \qquad s_{n+1} = (s_n, y_{n+1}). \tag{3}$$

This is a Markov chain on the growing space of contexts. Truncation rules such as top-k or nucleus sampling change π_T but not the structure.

The three maps are nested. Evaluating $z(s_n; \theta)$ in equation 3 means running equation 1 to the end, and computing the gradient in equation 2 means running equation 1 on teacher-forced contexts. The weights produced by equation 2 define the maps in equation 1 and equation 3. In practice the clocks also run at very different speeds relative to each other: generation never changes θ, and a single training step changes the forward map only by $O(\eta)$. In the language of slow-fast systems, clock 2 is the slow variable and clocks 1 and 3 are fast dynamics evaluated at a frozen slow state (Strogatz, 2018). Two well-studied effects break this picture: in-context learning, where the fast clock 1 appears to run an optimizer of its own (von Oswald et al., 2022; Akyürek et al., 2022), and self-training on model outputs, where clock 3 feeds clock 2 (Shumailov et al., 2023; Bertrand et al., 2023). Both are discussed in Sections 3 and 5.

2.2 A short primer

This subsection collects the dynamical-systems vocabulary used later. Readers who know it can skip to Section 3; a longer treatment is in Strogatz (2018).

Maps, fixed points and stability. A discrete-time system is a map $u_{k+1} = F(u_k)$. A fixed point $u^* = F(u^*)$ is linearly stable if every eigenvalue of the Jacobian $DF(u^*)$ lies inside the unit circle. For gradient descent on a quadratic with Hessian eigenvalue λ, the map $u \mapsto (1 - \eta\lambda)u$ is stable exactly when $\eta\lambda < 2$. This one-line calculation is the origin of the threshold $2/\eta$ in the edge-of-stability literature (Section 4.1).

Periodic orbits and bifurcations. A period-k orbit is a fixed point of F^k, and under a change of parameter such orbits can appear, disappear or exchange stability. A saddle-node bifurcation creates or destroys a pair of fixed points; after it, the system passes slowly through the region where they used to be (a "ghost"), which produces long plateaus followed by fast jumps (Strogatz, 2018; Dinc et al., 2025). In a period-doubling bifurcation an eigenvalue crosses -1 and a period-2 orbit is born; a cascade of such doublings is a standard route to chaos.

Lyapunov exponents. For a trajectory $u_0, u_1, \ldots$ the finite-time Lyapunov exponent over K steps is

$$\lambda_K(u_0) = \frac{1}{K} \log \sigma_{\max}\big(DF(u_{K-1}) \cdots DF(u_0)\big), \tag{4}$$

where $\sigma_{\max}$ is the largest singular value. Positive values mean that nearby trajectories separate exponentially; the limit $K \to \infty$, when it exists, is the largest Lyapunov exponent. Deep networks have a fixed and small number of steps, so only finite-time exponents make sense for clock 1 (Storm et al., 2023).

Basins and metastability. Boundaries between basins of attraction can be fractal, as in the hyperparameter space of neural network training (Sohl-Dickstein, 2024). A metastable state is one that is not an attractor but holds trajectories for a very long time, for example a cluster configuration that eventually merges (Geshkovski et al., 2024).

Noise-driven escape. Add noise of intensity D to a system with a stable state inside a well of depth ΔU. The mean time to leave the well grows as $\exp(\Delta U/D)$, with a prefactor set by the curvature at the bottom and at the barrier (Kramers, 1940; Hänggi et al., 1990). Escape from potential wells under deterministic periodic forcing is a related problem in mechanics, where the escape threshold in forcing amplitude depends on frequency and on the shape of the well (Farid & Gendelman, 2019; Farid, 2022). We use this lens for SGD in Section 4.2 and for sampling in Section 5.2.

Synchronization and consensus. Interacting units whose coupling pulls them together can reach a common state. Attention-based token dynamics over depth have this structure (Geshkovski et al., 2023b), and in two dimensions some of them reduce to Kuramoto-type models (Li et al., 2026).

Koopman linearization. A nonlinear map acts linearly on observables, $g \mapsto g \circ F$. Finite approximations of this Koopman operator give linear models of nonlinear dynamics that can be learned from data (Lusch et al., 2017; Brunton et al., 2015), and they have been applied to the training dynamics of neural networks (Dogra & Redman, 2020; Redman et al., 2021).

3 Clock 1: depth as time

3.1 Residual networks as discretized flows

Equation equation 1 is a forward Euler step of an ODE $\dot{x} = f(x, \ell)$ with unit step. This observation was made early for residual networks (He et al., 2015) by E (2017), Haber & Ruthotto (2017) and Lu et al. (2017), and it led to continuous-depth models (Chen et al., 2018), to architectures designed from the stability theory of ODEs and PDEs (Haber & Ruthotto, 2017; Ruthotto & Haber, 2018; Chang et al., 2019), to invertible second-order variants (Sander et al., 2021a), and to a view of training as optimal control (E et al., 2018). Whether a trained residual network actually converges to an ODE as depth grows depends on the initialization and on how the residual branch is scaled (Sander et al., 2022). In transformers, several stabilization methods amount to a choice of this step size, either by scaling the residual branch at initialization (Bachlechner et al., 2020; Wang et al., 2022) or by moving the layer normalization inside the branch (Xiong et al., 2020; Liu et al., 2020).

3.2 Attention as an interacting particle system

The transformer differs from a plain residual network because the update of token i depends on all tokens. Lu et al. (2019) read a transformer layer as a splitting scheme for a convection-diffusion equation of interacting particles. Sander et al. (2021b) showed that attention with doubly stochastic normalization is a Wasserstein gradient flow. Geshkovski et al. (2023a;b) studied a model in which tokens live on the unit sphere (a stand-in for layer normalization) and follow

$$\dot{x}_i = P_{x_i}\Big(\frac{1}{Z_i}\sum_j e^{\beta\langle Qx_i, Kx_j\rangle} V x_j\Big), \tag{5}$$

where P_x projects onto the tangent space and β acts as an inverse temperature. They proved that, for broad choices of Q, K, V, the tokens cluster as time (depth) grows, in the simplest cases to a single point. Clustering is the particle-system version of an older observation: stacks of pure attention lose rank doubly exponentially with depth (Dong et al., 2021), and rank collapse at initialization slows training (Noci et al., 2022). Similar results hold for oversmoothing in attention-based graph networks, which Wu et al. (2023) analyze as nonlinear time-varying dynamical systems.

Later work moved the model closer to real decoders. With causal masking the particle system is no longer a mean-field gradient flow, but clustering results still hold, and the first token, which attends only to itself, plays a special role (Karagodin et al., 2024). Wu et al. (2024) showed that the attention mask and layer normalization decide whether collapse happens at all. A mean-field limit with a Vlasov-type "Transformer PDE" covers several attention variants (Castin et al., 2025). Feed-forward layers can be read as a control that steers the particles (Maranzatto et al., 2026). Finite-width transformers with MLP blocks converge to a stochastic interacting particle system, in which noise can synchronize the tokens (Agazzi et al., 2026). For symmetric weights a spectral mode-selection mechanism decides the long-time state (Kuehn & Yoon, 2026), and in two dimensions deep linear transformers reduce to Kuramoto-type models with a rich set of behaviors (Li et al., 2026). The kinetic theory of causal attention has also been used to explain why models attend poorly to the middle of long contexts (Duerinckx et al., 2026; Liu et al., 2023).

Real models have finite depth, so the long-time limit of equation 5 describes them only partly. Geshkovski et al. (2024) proved that particles stay near a configuration of several clusters for an exponentially long time before they merge, and a stack of finite depth spends its layers inside such a long-lived transient.

3.3 Signal propagation, order and chaos

A second line of work treats a randomly initialized deep network as a random dynamical system and asks whether small input differences grow or shrink with depth. Poole et al. (2016) found an order-to-chaos transition controlled by the weight and bias variances; Schoenholz et al. (2016) showed that the depth over which signals propagate diverges at the transition, so networks are trainable to large depth only near it. The idea goes back to the chaotic phase of random recurrent networks (Sompolinsky et al., 1988). Requiring all singular values of the input-output Jacobian to be close to one (dynamical isometry) gives faster training still (Pennington et al., 2017) and allowed 10,000-layer convolutional networks to train (Xiao et al., 2018). For transformers, Cowsik et al. (2024) track the geometry of n token representations as a discrete-time system of interacting particles and derive conditions on the initialization that make deep transformers trainable.

For trained networks, Storm et al. (2023) computed finite-time Lyapunov exponents equation 4 of fully connected classifiers and found ridges of large exponents in input space that separate the classes, much like coherent structures in fluid flows. Our first experiment (Section 7.1) applies the same quantity to the residual stream of language models across training.

3.4 Trajectories in trained language models

Interpretability research gives a complementary, empirical picture of the depth dynamics in trained models. Predictions decoded from intermediate layers improve gradually, which suggests iterative refinement (Belrose et al., 2023). Deleting or swapping middle layers hurts much less than touching the first or last ones, consistent

with distinct stages of inference (Lad et al., 2024). The intrinsic dimension of hidden representations rises and then falls across layers (Valeriani et al., 2023). Token trajectories through depth cluster along a low-dimensional manifold and can be modeled as a stochastic process with few parameters (Sarfati et al., 2024). Fernando & Guitchounts (2025) treat the residual stream with tools from neural population dynamics and report that individual units change smoothly from layer to layer; Fernando & Guitchounts (2026) compute full Jacobian eigendecompositions of each layer in production-scale models. A small number of activations are orders of magnitude larger than the rest and nearly independent of the input (Sun et al., 2024); together with attention sinks (Xiao et al., 2023), they can be read as a near fixed point of the depth dynamics. Methods that read features off the layer Jacobian have also been analyzed theoretically (Ostrow et al., 2026). Recurrent-depth or looped models make the iteration explicit by applying the same block repeatedly, and their stability can be studied and scaled as such (Prairie et al., 2026).

3.5 When depth imitates training

Clock 1 can emulate clock 2. For in-context regression, a linear self-attention layer can implement one step of gradient descent on the in-context examples, so a stack of layers runs an optimizer whose iterations are the layers (von Oswald et al., 2022; Akyürek et al., 2022; Garg et al., 2022); trained linear transformers learn preconditioned versions of this algorithm (Ahn et al., 2023), and the gradient-flow dynamics that produce them can be solved in closed form (Zhang et al., 2023b). In this regime layer time is optimization time, and the step-size and stability questions of clock 2 carry over to clock 1.

3.6 Lessons from recurrent networks

Recurrent networks are the older case in which clocks 1 and 3 coincide: the same map is applied at every time step. Their analysis set many of the patterns used later for transformers. Fixed points and slow points of trained networks explain how they compute (Sussillo & Barak, 2013), and sentiment classifiers turn out to integrate evidence along a line attractor (Maheswaranathan et al., 2019). On the training side, exploding gradients were linked to bifurcations of the network dynamics (Pascanu et al., 2012; Ribeiro et al., 2019), and networks trained to reproduce chaotic dynamics must have diverging gradients (Mikhaeil et al., 2021). Lyapunov spectra quantify how much information a recurrent network keeps (Engelken et al., 2020; Vogt et al., 2020).

3.7 Explicit dynamical models of sequences

State-space models write the sequence dynamics in clock 3 explicitly, as a linear dynamical system in a hidden state, with eigenvalues placed to control memory and stability (Gu et al., 2021; Orvieto et al., 2023), or with input-dependent transitions (Gu & Dao, 2023). Linear attention has the same structure: it is a recurrent update of a matrix-valued state (Katharopoulos et al., 2020; Schlag et al., 2021). Recurrent networks built from coupled oscillators or constrained by Lipschitz bounds use the same design logic (Rusch & Mishra, 2020; Erichson et al., 2020). In these models the attractor and stability questions of this paper are explicit, since the eigenvalues of the state transition are parameters of the architecture.

4 Clock 2: training as a dynamical system

4.1 Gradient descent at the edge of stability

Classical optimization theory treats gradient descent as a discretization of gradient flow, which is valid when the step size is small compared with the inverse curvature. Cohen et al. (2021) showed that neural network training does not stay in that regime. Under full-batch gradient descent the sharpness, meaning the largest Hessian eigenvalue $\lambda_{\max}$, first rises (progressive sharpening) until it reaches $2/\eta$, and then hovers near that value while the loss keeps decreasing in a non-monotone way. By the linear stability argument of Section 2.2, the iteration then sits on the boundary where the top direction loses stability. Adaptive methods show the same behavior with respect to the preconditioned sharpness (Cohen et al., 2022).

Several mechanisms have been proposed for why training survives at this boundary. Damian et al. (2022) show that the third-order term of the loss creates a negative feedback: when the iterate oscillates along the top eigenvector, the oscillation itself reduces the sharpness. Arora et al. (2022) and Ahn et al. (2022) give conditions under which gradient descent converges while being locally unstable. Minimal models help because they can be solved: Zhu et al. (2022) find edge-of-stability dynamics in a two-parameter example, and Chen & Bruna (2022) show that beyond the threshold gradient descent can converge to stable period-two orbits, a period-doubling bifurcation in the strict sense. Backward error analysis of the finite step shows that gradient descent follows a modified flow with an implicit penalty on the gradient norm (Barrett & Dherin, 2020).

Larger step sizes push the map further. Lewkowycz et al. (2020) identified a catapult phase in which the loss first grows, the curvature drops, and training settles in a flatter region; above a second threshold training diverges. Kalra & Barkeshli (2023) mapped the early-training regimes as a function of learning rate, depth and width. Kong & Tao (2020) showed that deterministic gradient descent on a multiscale objective, with a step size that resolves only the macroscopic scale, can become chaotic and converge to a statistical distribution. Tuci et al. (2026) model stochastic optimizers at the edge of stability as random dynamical systems with fractal attractors and relate the attractor dimension to generalization. Sohl-Dickstein (2024) found that the boundary in hyperparameter space between step sizes that train and those that diverge is fractal, in the same way as the boundary of bounded orbits of the quadratic map that defines the Mandelbrot set. Mini-batch sampling moves this stability boundary even when gradients are unbiased, as shown exactly for linear self-attention (Balasubramanian, 2026). Curvature control has been used to keep large transformers away from the boundary (Ramasinghe et al., 2026; Gilmer et al., 2021).

4.2 Stochastic gradient descent as a noisy dynamical system

Minibatch noise turns clock 2 into a stochastic process. Stochastic modified equations approximate SGD by an SDE whose diffusion is set by the gradient covariance and the step size (Li et al., 2015). Near a minimum the process behaves like an Ornstein-Uhlenbeck process, with the ratio of step size to batch size acting as a temperature (Mandt et al., 2017; Smith & Le, 2017; Jastrzębski et al., 2017). Because SGD noise is anisotropic and state dependent, the stationary distribution is not a Gibbs distribution of the training loss, and the dynamics can converge to limit cycles instead of points (Chaudhari & Soatto, 2017).

Escape from a minimum is the most direct use of the escape theory sketched in Section 2.2. Xie et al. (2020) derived a Kramers-type rate for SGD and found that, unlike isotropic Langevin dynamics, SGD favors flat minima exponentially in the ratio of barrier height to Hessian eigenvalues. Anisotropic noise aligned with the curvature helps escape from sharp minima (Zhu et al., 2018). Mori et al. (2021) showed that the multiplicative character of the noise changes the law: the escape rate depends on the logarithm of the loss barrier, which gives an escape rate that scales as a power of the loss ratio between saddle and minimum. Heavy-tailed gradient noise points toward Lévy-driven escape, which is dominated by single large jumps (Simsekli et al., 2019). The precise law therefore depends on the noise model, and the law that holds for language model pretraining has not been measured.

4.3 Saddles, plateaus and ghosts

High-dimensional loss surfaces have far more saddles than local minima (Dauphin et al., 2014). Noise or small perturbations let gradient methods leave strict saddles efficiently (Ge et al., 2015; Jin et al., 2017), but the time spent near a saddle can still dominate training. For deep linear networks the learning dynamics can be solved exactly: modes of the input-output map are learned one after another, separated by plateaus whose length depends on the initialization (Saxe et al., 2013). Spectral bias and the frequency principle describe a similar ordering, with low-frequency components learned first (Rahaman et al., 2018; Xu et al., 2019). In recurrent networks, abrupt learning after a long plateau has been traced to the ghost of a saddle-node bifurcation (Dinc et al., 2025), and learning can itself drive a network from chaotic to stable activity (Vaidya, 2026).

4.4 Abrupt transitions in what the model can do

Grokking, where test accuracy jumps long after the training loss has saturated, was first reported on algorithmic data (Power et al., 2022). Inside the network the change is gradual: circuit-level measures improve steadily before the jump (Nanda et al., 2023). Explanations include an effective theory of representation learning with a phase diagram (Liu et al., 2022a), a dependence on weight norm (Liu et al., 2022b), a transition from lazy to rich training (Kumar et al., 2023), two phases of implicit bias (Lyu et al., 2023), and a first-order phase transition in a solvable two-layer model (Rubin et al., 2023). With adaptive optimizers, grokking coincides with cyclic phase changes and loss spikes (Thilak et al., 2022).

Language models show similar transitions. Induction heads form during a short window of training, and the loss curve shows a visible bump at that moment (Olsson et al., 2022). Syntactic attention structure appears abruptly and is followed by a drop in loss (Chen et al., 2023). In-context learning can emerge and later fade while training continues (Singh et al., 2023); minimal models explain the abruptness (Reddy, 2023), and Bayesian models of attention predict phase transitions in the emergence of copy heads (Lavie et al., 2026). Singular learning theory relates these stages to changes in the degeneracy of the loss landscape (Hoogland et al., 2024). Across model scale, abilities can appear suddenly (Wei et al., 2022), although part of the effect comes from discontinuous metrics (Schaeffer et al., 2023); one model attributes emergence to discrete skills that are learned in order of frequency (Michaud et al., 2023).

4.5 Loss spikes in large-scale pretraining

Large pretraining runs show sudden loss spikes. PaLM handled them by restarting from an earlier checkpoint and skipping data (Chowdhery et al., 2022). Wortsman et al. (2023) reproduced two instability mechanisms at small scale, growth of attention logits and divergence of the output logits, and studied how they depend on learning rate. Attention entropy collapse is another proposed cause (Zhai et al., 2023), as is the behavior of the Adam update at large scale (Molybog et al., 2023); conditions on the scale of embeddings and residual connections can prevent spikes (Takase et al., 2023). From the dynamical point of view these are instabilities of the map equation 2 near the edge of stability, and the open question is which kind (Section 8, **OP 4**).

4.6 Slow phases, landscape geometry and conserved quantities

Pretraining curves typically show a fast initial drop followed by a long slow phase. Zhang et al. (2026a) observe that the normalized singular value spectrum of the weight matrices stabilizes early and that this stabilization coincides with the slow phase. The river-valley picture explains why a constant learning rate followed by a fast decay works: the iterate moves along a valley floor while oscillating across it (Wen et al., 2024). The Hessian has a bulk of near-zero eigenvalues and a few outliers (Sagun et al., 2017; Ghorbani et al., 2019), and the gradient lies mostly in the subspace of the outliers (Gur-Ari et al., 2018). Two copies of a run that are split at step k and then see different minibatch noise end in the same linearly connected region only when k is past a short early phase; before that, the trajectory is sensitive to small perturbations (Frankle et al., 2019). Continuous symmetries of the architecture imply conservation laws for gradient flow, which finite steps and noise break in predictable ways (Kunin et al., 2020). At the end of classification training, features converge to a highly symmetric configuration (Papyan et al., 2020).

Infinite-width limits give two simpler dynamical systems. In the lazy limit, training is linear in function space (Jacot et al., 2018). In the mean-field limit, training two-layer networks is a Wasserstein gradient flow of particles (Mei et al., 2018; Chizat & Bach, 2018), and parametrizations exist that keep feature learning at infinite width (Yang & Hu, 2020). Scaling laws describe the slow phase at the level of loss averaged over the whole run (Kaplan et al., 2020; Hoffmann et al., 2022). Koopman operator methods have been used to speed up training and to explain magnitude pruning (Dogra & Redman, 2020; Redman et al., 2021).

5 Clock 3: generation as a dynamical system

5.1 Decoding as a stochastic map

Definition 3 makes generation a Markov chain whose state is the whole context. The state space grows with n, so the usual results on stationary distributions do not apply directly, but the questions of dynamical systems still make sense: which sets of contexts trap the chain, how long the chain stays in them, and how these answers depend on the temperature T and on truncation. Krishnamurthy et al. (2026) formulate language models as high-dimensional nonlinear autoregressive models and analyze networks of interacting models with mean-field equations. A deterministic version of clock 3, obtained by feeding back the expected embedding of the next-token distribution instead of a sample, is a continuous map that can be analyzed with standard tools (Jones, 2026).

Temperature divides the logits by T, so π_T in equation 3 is a Gibbs distribution with energy $-z$ at temperature T. Greedy decoding is the limit $T \to 0$, and T sets how often the chain leaves the most likely path, so it serves as both the noise intensity and the natural bifurcation parameter of clock 3. Its effect on task accuracy (Renze & Guven, 2024) and on novelty and coherence (Peeperkorn et al., 2024) has been measured directly.

5.2 Repetition loops as attractors

With greedy or low-temperature decoding, models fall into loops that repeat a phrase or sentence until the length limit (Holtzman et al., 2019). At $T = 0$ the chain is deterministic and the loop is one of its periodic orbits. Xu et al. (2022) showed that loops reinforce themselves: the probability of repeating a sentence grows with the number of times it already appears in the context. Unlikelihood training (Welleck et al., 2019), truncation rules (Hewitt et al., 2022; Meister et al., 2022) and feedback control of perplexity (Basu et al., 2020) are partly attempts to keep the chain out of these attractors; Fu et al. (2020) give a theoretical analysis of why they form. Recent work studies the inside of the loop: how it shows up in residual-stream dynamics before it becomes visible in the output (Zhang et al., 2026b), and the conditions under which a loop "locks" so that escape becomes impossible in practice under a given sampler (Chen & Wang, 2026). Loops also appear at the level of whole texts. Repeated paraphrasing converges to period-two attractor cycles even with sampling noise (Wang et al., 2025).

Escape from a loop has the structure of the noise-driven escape problem of Section 2.2. Suppose the chain has locked into a loop of period k with tokens $a_1, \dots, a_k$. As long as the sampler keeps choosing the loop token, the context is deterministic, so the logits z_t along the loop are fixed numbers. Define the logit gaps $\Delta_t^j = z_t(a_t) - z_t(j)$ for $j \neq a_t$; all are positive because the loop is the greedy path. The probability of staying in the loop at step t is

$$p_t(T) = \frac{1}{1 + S_t(T)}, \qquad S_t(T) = \sum_{j \neq a_t} e^{-\Delta_t^j / T}, \tag{6}$$

and the probability of leaving during one cycle is $r(T) = 1 - \prod_{t=1}^{k} p_t(T)$.

Proposition 1 (Arrhenius law for loop escape)**.** *Let $\Delta_{\min} = \min_t \min_{j \neq a_t} \Delta_t^j$ and let m be the number of pairs (t, j) that attain it. Then $r(T) \leq \sum_t S_t(T)$ for all $T > 0$, and*

$$r(T) = m\, e^{-\Delta_{\min}/T} \left(1 + o(1)\right) \quad \text{as } T \to 0, \qquad \text{so} \qquad \lim_{T \to 0} T \log r(T) = -\Delta_{\min}. \tag{7}$$

Proof. Since $1 - \prod_t (1 - q_t) \leq \sum_t q_t$ for $q_t \in [0, 1]$, and $1 - p_t = S_t/(1 + S_t) \leq S_t$, the bound follows. As $T \to 0$, each $S_t(T)$ is dominated by its smallest gaps, $S_t(T) = m_t e^{-\delta_t/T}(1 + o(1))$ with $\delta_t = \min_j \Delta_t^j$, and $1 - \prod_t p_t = \sum_t S_t + O(\max_t S_t^2)$. The terms with $\delta_t = \Delta_{\min}$ dominate the sum. □

The logit gap therefore plays the role of the barrier height ΔU in Kramers' formula (Kramers, 1940), and T plays the role of noise intensity. Proposition 1 follows almost directly from the Gibbs form of the softmax. The barrier, however, changes with time. Because loops reinforce themselves (Xu et al., 2022), the gaps

grow with each completed cycle, so the well deepens while the chain sits in it. Let Δ_c be the smallest gap during cycle c and $r_c(T)$ the escape probability during that cycle. The probability of never leaving is $\prod_{c\geq 1}(1 - r_c(T))$, which is positive exactly when $\sum_c r_c(T) < \infty$ (given $r_c < 1$).

Proposition 2 (Deepening wells lock)**.** *If $\Delta_c \geq \Delta_1 + b\,(c-1)$ for some $b > 0$, then for every $T > 0$ the probability of never escaping is at least $1 - k(|\mathcal{V}| - 1)\, e^{-\Delta_1/T}/(1 - e^{-b/T})$ whenever this bound is positive, and it is positive for every $T > 0$. If instead the logits along the loop converge from cycle to cycle, so that $\Delta_c \to \Delta_\infty < \infty$, escape eventually happens with probability one, but the expected number of cycles before escape grows like $e^{\Delta_\infty/T}$ as $T \to 0$.*

Proof. By the bound in Proposition 1, $r_c \leq k(|\mathcal{V}| - 1)e^{-\Delta_c/T}$, which is summable under linear growth, and $\prod_c(1 - r_c) \geq 1 - \sum_c r_c$. If the logits converge, then r_c tends to a positive limit and the second statement follows from the first part of Proposition 1. □

In practice logits are bounded, so the gaps must saturate and the second case applies; "locking" then means that the escape time exceeds any realistic generation budget. Experiment 3 (Section 7.3) measures Δ_c along real loops of GPT-2 and tests both propositions against the exact survival probabilities and against Monte Carlo sampling.

5.3 Temperature-driven transitions

At the level of statistics over many generations, temperature produces something close to a phase transition. Nakaishi et al. (2024) varied T in GPT-2 and found a low-temperature phase with repetitive structure, a high-temperature phase of incoherent text, and power-law correlations close to those of natural language at the transition point. Arnold et al. (2024) proposed statistical methods to detect such transitions automatically from output distributions. Jones (2026) found that in the continuous feedback version of generation, temperature and activation steering move the system between periodic orbits and chaos with positive finite-time Lyapunov exponents, with period-doubling scaling and periodic orbits proved by interval arithmetic. In chain-of-thought trajectories, Jaca et al. (2026) report sensitive dependence on initial conditions together with bounded evolution and fractal structure in hidden-state space. Sorting many generated trajectories into regimes (fixed point, periodic, irregular) without labels is a response-regime identification problem, for which unsupervised data-driven methods exist in mechanics (Farid, 2024).

5.4 Drift over long horizons

Training uses teacher forcing, while generation feeds the model its own outputs. Small errors therefore compound along clock 3, a problem known as exposure bias (Bengio et al., 2015; Wang & Sennrich, 2020). Errors snowball in multi-step reasoning (Dziri et al., 2023) and in hallucinated justifications (Zhang et al., 2023a), and next-token prediction can fail on planning tasks that require looking ahead (Bachmann & Nagarajan, 2024). In dynamical terms these are statements about the growth of perturbations along a trajectory of the closed-loop system, which is what Lyapunov exponents measure.

5.5 Outer loops: when generation feeds training

The slow-fast separation of Section 2.1 breaks when generated text is used for training. Training repeatedly on a model's own outputs makes the tails of the distribution disappear (Shumailov et al., 2023; Alemohammad et al., 2023); whether the process stays near the original model depends on the fraction of real data in each round (Bertrand et al., 2023). Reinforcement learning from human feedback reduces output diversity (Kirk et al., 2023), which can be read as a contraction of the set of trajectories that clock 3 visits. These are dynamical systems over model versions, a fourth and much slower clock.

5.6 Language models as models of dynamical systems

A different line of work asks whether language models can forecast dynamical systems placed in their context. Pretrained models forecast time series zero-shot (Gruver et al., 2023) and extrapolate trajectories of physical

systems with accuracy that improves with context length (Liu et al., 2024). Chaotic systems serve as a benchmark for such forecasters (Gilpin, 2021; 2023; Pathak et al., 2018), and training on data from cellular automata at the edge of chaos produces models with better downstream reasoning (Zhang et al., 2024). These results are about the content a model processes, not about the model as a dynamical system, but they share tools with this paper.

6 The Three-Clock Dictionary

Table 1 collects the results of Sections 3, 4 and 5. Each row is a concept from nonlinear dynamics and each column is a clock. A cell names the observed behavior that the concept describes on that clock and marks its evidence level: P when the correspondence is proved for a model of the system, M when it has been measured in trained networks, and C when it is a conjecture. A question mark means that we found no work that tests the concept on that clock.

Reading the table by rows. Fixed points, stability and Lyapunov exponents have been used on all three clocks, although with different meanings: on clock 1 they describe a single finite trajectory, on clock 2 the long-run behavior of an optimizer, and on clock 3 a random process. Phase transitions and escape appear on clocks 2 and 3 but not on clock 1, where the number of steps is fixed and small, so there is no long-time limit in which a transition could be sharp. Even so, the metastability result of Geshkovski et al. (2024) shows that finite-time transients on clock 1 can be analyzed with care.

Reading the table by columns. Clock 2 is the best covered column, because optimization theory has always been a theory of iterated maps. Clock 1 is covered mostly by results at initialization and by models with simplified attention. Clock 3 is the youngest column, and most of its entries date from 2024 or later.

Where the analogies are weak. First, clock 1 runs for a fixed and small number of steps (between 6 and about 100 layers), so asymptotic notions such as attractors and Lyapunov exponents must be replaced by finite-time versions. Second, clock 2 is driven by a data stream whose statistics are not stationary in general, and by learning-rate schedules, which makes it a non-autonomous system. Third, the state of clock 3 is the whole context, so its dimension grows with time and the chain never returns to an earlier state. Statements about loops and attractors on clock 3 are therefore about the projection of the state onto recent tokens. Each of these differences limits which theorems can be imported from the classical theory without change.

7 Three experiments, one per clock

Each experiment below takes one concept from Table 1, measures it on open models, and runs on a two-core CPU in under two hours. Details are in Appendix A, and the code reproduces every number.

7.1 Experiment 1: the Lyapunov profile of the residual stream over training (clocks 1 and 2)

Question and setup. Finite-time Lyapunov exponents equation 4 describe how a forward pass stretches perturbations, layer by layer. How does this profile change as the model trains? This question couples clock 1 to clock 2 (**OP 1**). We use the public training checkpoints of Pythia (Biderman et al., 2023): eleven checkpoints of Pythia-70M (6 layers), from initialization to step 143,000, and six checkpoints of Pythia-160M (12 layers). Each block is treated as a map F_ℓ on the full residual stream of a 16-token context. For each block we estimate the largest singular value σ_ℓ of its Jacobian by power iteration and report the relative exponent

$$\tilde{\lambda}_\ell = \log \sigma_\ell - \log \left(\|x^{\ell+1}\| / \|x^\ell\| \right), \tag{8}$$

which measures stretching beyond the growth of the residual norm. For the whole stack we report $\Lambda = \frac{1}{L} \log \sigma_{\max}(DF_{0\to L})$, and the ratio $\kappa = \log \sigma_{\max}(DF_{0\to L}) / \sum_\ell \log \sigma_\ell$. Since $\sigma_{\max}$ of a product is at most the product of the σ_ℓ, $\kappa \le 1$ whenever $\sum_\ell \log \sigma_\ell > 0$, which holds throughout our measurements, and κ near one means that the most expanded direction of each layer is passed on to the next. Values are means over eight prompts, and κ is the ratio of the two means.

Concept	Clock 1: layer time ℓ	Clock 2: training time t	Clock 3: generation time n
Fixed point	Tokens collapse to a cluster; rank collapse P (Geshkovski et al., 2023b; Dong et al., 2021). Massive activations nearly input independent M (Sun et al., 2024)	Convergence to a minimum; neural collapse at the end of training M (Papyan et al., 2020)	Period-one loops (one token repeated) as the simplest case of degenerate repetition M (Holtzman et al., 2019)
Linear stability, Jacobian spectrum	Dynamical isometry; layer Jacobians of trained models P/M (Pennington et al., 2017; Fernando & Guitchounts, 2026)	Sharpness vs. $2/\eta$ M (Cohen et al., 2021)	Derivative of the feedback map factorizes through $1/T$ M (Jones, 2026)
Finite-time Lyapunov exponent	Ridges in input space separate classes M (Storm et al., 2023); per-layer profile over training M (Sec. 7.1)	Sensitivity of runs to minibatch noise early in training M (Frankle et al., 2019)	Divergence of nearby generations M (Jaca et al., 2026; Jones, 2026)
Edge of chaos, criticality	Order-chaos transition at initialization sets trainable depth P (Poole et al., 2016; Schoenholz et al., 2016)	Edge of stability M (Cohen et al., 2021)	Critical temperature with power-law text statistics M (Nakaishi et al., 2024)
Limit cycle (periodic orbit)	?	Period-2 orbits of GD P (Chen & Bruna, 2022); SGD converges to cycles P (Chaudhari & Soatto, 2017)	Repetition loops M (Xu et al., 2022); paraphrase 2-cycles M (Wang et al., 2025)
Period doubling, route to chaos	?	Larger step sizes give period doubling and chaos in minimal models P (Zhu et al., 2022; Kong & Tao, 2020); period-2 orbit, then irregular oscillation, in a transformer M (Exp. 2)	Feigenbaum scaling in embedding feedback M (Jones, 2026)
Fractal boundary	?	Trainable vs. divergent hyperparameters M (Sohl-Dickstein, 2024); interleaved trainable and divergent step sizes M (Exp. 2); fractal attractors of optimizers M (Tuci et al., 2026)	?
Basin of attraction	Class regions bounded by FTLE ridges M (Storm et al., 2023)	Which minimum is reached depends on step size and batch size M (Jastrzębski et al., 2017)	Prompts that do or do not fall into loops C
Metastability, slow manifold	Multi-cluster states last exponentially long P (Geshkovski et al., 2024)	Plateaus P (Saxe et al., 2013); slow phase of pretraining M (Zhang et al., 2026a; Wen et al., 2024)	Long coherent stretches before drift C
Saddle and ghost	?	Saddles dominate M (Dauphin et al., 2014); abrupt learning from a ghost P (Dinc et al., 2025)	?
Phase transition	?	Grokking P/M (Rubin et al., 2023; Power et al., 2022); induction heads M (Olsson et al., 2022)	Low-T repetitive vs. high-T incoherent phase M (Nakaishi et al., 2024; Arnold et al., 2024)
Noise-driven (Kramers) escape	?	Escape of SGD from minima P (Xie et al., 2020; Mori et al., 2021)	Escape from repetition loops, logit gap as barrier P/M (Prop. 1, Exp. 3)
Synchronization, consensus	Clustering as consensus; Kuramoto form P (Geshkovski et al., 2023b; Li et al., 2026); synchronization by noise P (Agazzi et al., 2026)	?	Collective dynamics of interacting models P (Krishnamurthy et al., 2026)
Koopman, linearization	Local linear description of each layer M (Fernando & Guitchounts, 2026)	Koopman-based training and pruning M (Dogra & Redman, 2020; Redman et al., 2021); lazy regime P (Jacot et al., 2018)	?
Conservation law, symmetry	Gradient-flow structure of attention P (Sander et al., 2021b)	Symmetries give conserved quantities that finite steps break P (Kunin et al., 2020)	?

Table 1: The Three-Clock Dictionary. Each cell gives the behavior that a dynamical-systems concept describes on one clock, with its evidence level (P: proved in a model; M: measured in trained networks; C: conjecture) and representative references. Question marks are cells for which we found no study; they define the open problems of Section 8.

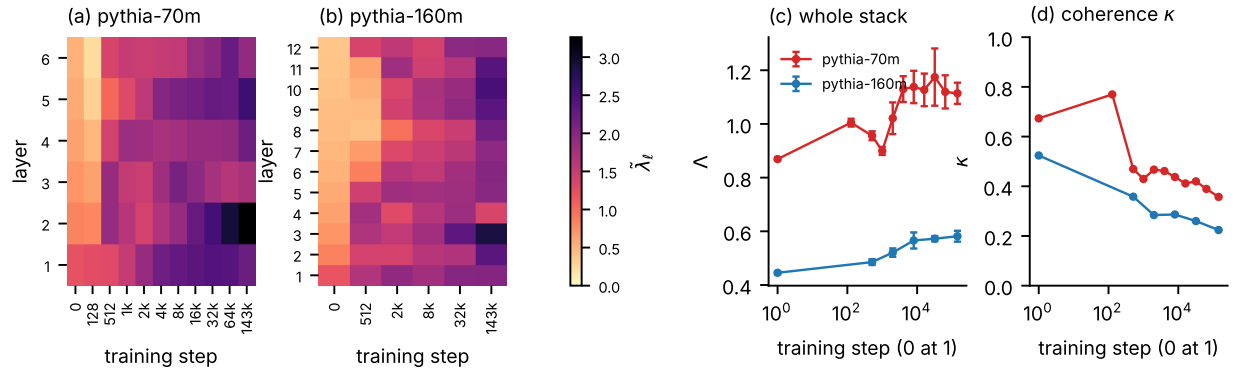


Figure 2: Experiment 1: finite-time Lyapunov exponents of the residual stream across public Pythia checkpoints. (a, b) Relative per-layer exponent $\tilde{\lambda}_\ell$ (stretching of the top singular direction beyond the growth of the residual norm) for each layer and checkpoint. (c) Depth-averaged exponent of the whole stack, Λ; bars: standard deviation over prompts. (d) Coherence κ between per-layer and whole-stack stretching; values below one mean that the most expanded directions of consecutive layers are not aligned.

Results. Figure 2 shows three things. First, at initialization both models have the same profile, decreasing with depth ($\tilde{\lambda}_\ell$ from 1.20 to 0.53 in Pythia-70M and from 1.19 to 0.39 in Pythia-160M), which reflects the shared initialization scheme. Second, training makes every layer more expansive. The mean of $\tilde{\lambda}_\ell$ over layers rises from 0.75 to 2.32 in Pythia-70M and from 0.59 to 2.17 in Pythia-160M, and it is still rising at the last checkpoint. The rise is not monotone at the start, since at step 128 the deepest layers of Pythia-70M contract further (layer 6 drops from 0.53 to 0.22) before all layers expand. Third, the whole stack does not follow. In Pythia-70M, Λ goes from 0.87 to about 1.1 and stays between 1.11 and 1.18 from step 4,000 onward, which is under 3% of training. In Pythia-160M it moves from 0.45 to 0.58. As a result the coherence κ falls during training, from 0.67 to 0.36 in Pythia-70M and from 0.52 to 0.22 in Pythia-160M.

Reading. Training produces layers that are individually more sensitive to perturbations, while the expanding directions of consecutive layers become less aligned, so the forward pass as a whole stretches perturbations only moderately. The depth-averaged exponent settles early, while the per-layer exponents keep changing. This is consistent with the observation that the normalized singular value spectra of the weights stabilize early in pretraining (Zhang et al., 2026a), now seen through the Jacobians of the forward map. These are measurements on two small models with short contexts and a small prompt set. Whether the same pattern holds at scale, and whether κ predicts robustness to perturbations or the success of model editing, is part of **OP 1**.

7.2 Experiment 2: a learning-rate bifurcation diagram (clock 2)

Question and setup. How does the long-run behavior of gradient descent on a transformer change with the step size η? We train a one-block transformer (two heads, width 32) with full-batch gradient descent on next-token prediction of sequences from a fixed random Markov chain, for 56 step sizes between 0.02 and 4. Within one seed every step size starts from the same initialization and sees the same data, and we repeat the sweep for three seeds that change both. Full batches remove minibatch noise, so any irregularity comes from the map equation 2 itself. For each run we record the last 400 losses and the sharpness $\lambda_{\max}$ at the end.

Results. Figure 3 shows three regimes. In each seed, 48 of the 56 step sizes train. Across these 144 runs the final sharpness sits just above the stability threshold, with $\eta\lambda_{\max}/2$ having median 1.07 and range $[1.00, 1.21]$ across two decades of step size (panel b). This is the edge-of-stability behavior of Cohen et al. (2021), and in this model it holds over the entire trainable range we tested. None of the runs settles on a fixed point. Removing the slow downward trend of the loss shows what the iterate does instead (panels a

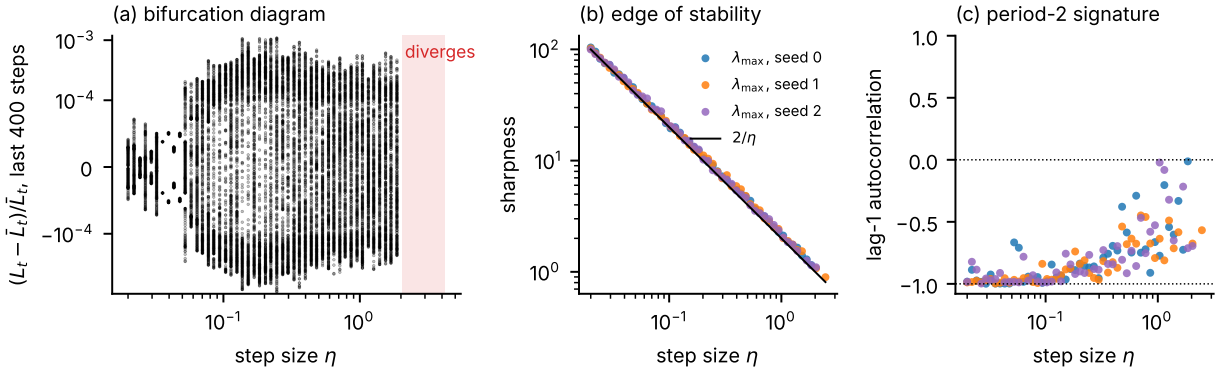


Figure 3: Experiment 2: full-batch gradient descent on a one-block transformer. (a) Bifurcation diagram for seed 0: relative deviation of the last 400 losses from their running mean (9-step window, symmetric-log axis) against the step size. Two branches at small η mark a period-2 orbit; the band at larger η is irregular oscillation; the shaded region diverges. (b) Sharpness at the end of training against $2/\eta$ for three seeds. (c) Lag-1 autocorrelation of the detrended loss for the same runs; -1 is a clean period-2 orbit.

and c). For $\eta < 0.05$ the fluctuations alternate in sign at every step, a period-2 orbit riding on the slow descent, and the median lag-1 autocorrelation of the detrended loss over the ten smallest step sizes is -0.99, -0.99 and -0.97 in the three seeds. As η grows the fluctuations become larger and less regular. The median lag-1 autocorrelation rises to between -0.95 and -0.92 for $0.05 \le \eta < 0.3$, between -0.74 and -0.68 for $0.3 \le \eta < 1$, and between -0.66 and -0.32 for $\eta \ge 1$. The edge of trainability depends on the seed. In seed 0 every $\eta \ge 2.04$ diverges, while in seeds 1 and 2 divergent and trainable step sizes interleave between 1.5 and 2.5, and in seed 1 the largest trainable step size (2.47) lies above the smallest divergent one (1.68).

Reading. In dictionary terms, the trainable range of this model is a sequence from a period-2 orbit near the stability boundary, through increasingly irregular oscillation, to divergence. This is the picture that minimal models predict (Chen & Bruna, 2022; Zhu et al., 2022; Kong & Tao, 2020), here measured on a transformer with attention and layer normalization. The interleaving of trainable and divergent step sizes near the edge is consistent with the fractal trainability boundary reported by Sohl-Dickstein (2024), although a grid of 56 points cannot measure a fractal dimension. We do not claim a full period-doubling cascade, since with 400-step tails on top of a slowly drifting loss we cannot resolve periods above two.

7.3 Experiment 3: escape from repetition loops (clock 3)

Question and setup. Section 5.2 predicts that escape from a repetition loop follows an Arrhenius law with the logit gap as barrier (Proposition 1), and that loops lock if the barrier grows from cycle to cycle (Proposition 2). We test both on GPT-2 small (Radford et al., 2019) and on the final checkpoint of Pythia-160M (Biderman et al., 2023). From each of 40 prompts we decode greedily for 320 tokens and keep the runs that end in a loop. Because the context along a loop is deterministic, the logits stored during the greedy run give the exact probability of staying in the loop at any temperature, and we check these exact values against Monte Carlo sampling.

Results. Greedy decoding ends in a loop for 37 of the 40 prompts in each model, with periods from 4 to 53 tokens (median 23) in GPT-2 and from 7 to 46 tokens (median 22) in Pythia-160M (Figure 4).

The barrier grows. The median smallest logit gap Δ_c rises from 0.36 in the second cycle of a loop to 2.33 in the sixth and 3.63 in the twentieth in GPT-2, and from 0.47 to 1.83 and then 2.77 by the tenth cycle in Pythia-160M (panel a). The barrier in cycle 6 is larger than in cycle 2 for 36 of 36 GPT-2 loops and 35 of 37 Pythia-160M loops. This is the self-reinforcement reported by Xu et al. (2022), measured here as the depth of a well.

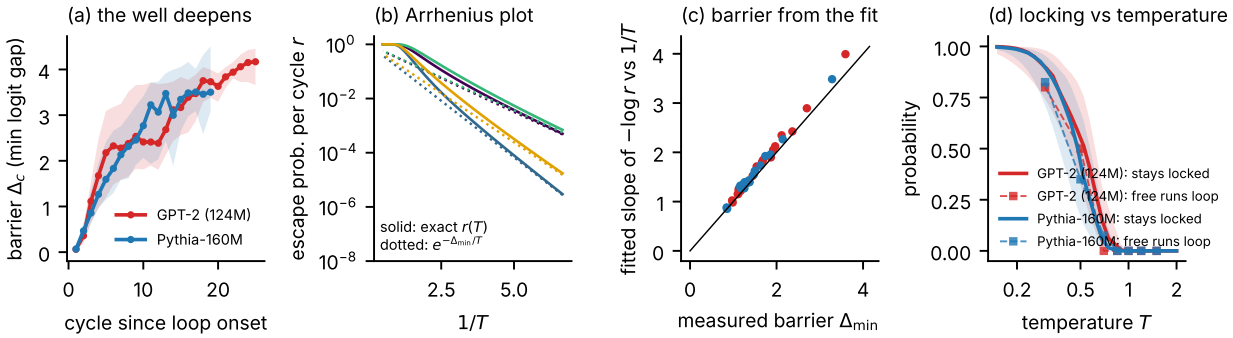


Figure 4: Experiment 3: escape from repetition loops. (a) Barrier Δ_c (smallest logit gap in cycle c) against cycles since loop onset; line: median over loops, band: interquartile range; only cycles reached by at least 10 loops are shown. (b) Exact escape probability per cycle for four GPT-2 loops (solid) and the Arrhenius term $e^{-\Delta_{\min}/T}$ (dotted). (c) Barrier recovered from the slope of $\log r$ against $1/T$ at low temperature, against the measured $\Delta_{\min}$; line: identity. (d) Solid: median probability of staying in the loop until the end of the 320-token horizon, after two cycles of burn-in (band: interquartile range); the number of remaining cycles differs between loops because their onsets differ. Dashed: fraction of free generations of 200 tokens that end in a loop.

Escape follows the Arrhenius law. For the first cycle after two cycles of burn-in, the escape probability per cycle approaches $e^{-\Delta_{\min}/T}$ at low temperature (panel b). Fitting $\log r$ against $1/T$ recovers the measured barrier with a correlation of 0.997 in both models and a median relative difference of 6.9% (22 GPT-2 loops) and 5.7% (21 Pythia-160M loops). Loops whose escape probability never falls below 10^{-2} on our temperature grid are excluded from the fit (panel c). The exact survival probabilities agree with Monte Carlo sampling. Across 24 comparisons per model (four loops, two temperatures, survival after 1, 2 and 4 cycles, 128 samples each), the largest absolute difference is 0.062 for GPT-2 and 0.068 for Pythia-160M, and every difference is within two binomial standard errors.

Loops lock below a fairly sharp temperature. The median probability of staying in a loop until the end of the 320-token horizon is 0.89 at $T = 0.31$ in both models, crosses 0.5 at $T \approx 0.53$ (GPT-2) and $T \approx 0.48$ (Pythia-160M), and falls to 0.06 and 0.015 at $T = 0.75$ (panel d). Free sampling gives a consistent picture. After 200 tokens, 80% (GPT-2) and 82.5% (Pythia-160M) of generations end in a loop at $T = 0.3$, 50% and 35% at $T = 0.5$, at most 7.5% at $T = 0.7$, and none at $T \geq 0.85$.

Reading. The escape problem on clock 3 has the structure of Kramers' problem, with two differences that matter. The noise acts on a discrete choice through a Gibbs distribution, so the Arrhenius form follows directly from the softmax (Proposition 1) and needs no diffusion approximation. In addition, the barrier is a function of the residence time, so a loop becomes harder to leave the longer the model repeats it. Panel (a) also shows that the deepening is not uniform, and barriers often rise quickly over the first five cycles, pause, and rise again. Escape from a well whose depth grows with the time already spent in it is a non-stationary escape problem. Its mechanical analogue is a particle in a well that deepens as the particle settles, and its scaling with model size is open (**OP 5**).

8 Open problems

Each problem below comes from an empty or weak cell of Table 1, or from a place where two clocks meet, and each is stated with a first test. Problems **OP 9** and **OP 10** are conjectures that come from the mechanics of nonlinear oscillators. We include them because they make predictions that can be checked, and we do not claim evidence for them yet.

OP 1. Coupling of clocks 1 and 2. How does training reshape the depth dynamics, and which features of the Lyapunov profile settle before the loss does? Experiment 1 gives a first answer for two small models, in which the whole-stack exponent settles within 3% of training while the per-layer exponents keep growing. The natural next test is the full Jacobian spectrum of every layer across the public checkpoints of a model family with more than a billion parameters.

OP 2. Periodic orbits in depth. Do models that reuse a block many times, such as looped and recurrent-depth models (Prairie et al., 2026), have periodic or quasi-periodic orbits in the residual stream, and do these orbits carry computation? Return maps of hidden states over many loop iterations would show such orbits directly.

OP 3. The full Lyapunov spectrum of a language model. Beyond the largest exponent, how many directions expand along a forward pass and along a generation, and what is the resulting attractor dimension? QR-based algorithms (Engelken et al., 2020; Vogt et al., 2020) make this computation feasible for models of GPT-2 size.

OP 4. The type of a loss spike. Are loss spikes in pretraining crossings of the stability boundary, period-doubling cascades, catapults (Lewkowycz et al., 2020), or escapes from a basin? Each type leaves a different signature in the curvature and in the update direction just before the spike, so spikes in open training logs and checkpoints can be classified by these signatures.

OP 5. Escape laws across samplers and scales. Proposition 1 fixes the low-temperature law for plain temperature sampling. How do nucleus sampling and repetition penalties change the barrier, and how does the growth of the barrier from cycle to cycle scale with model size? Experiment 3 can be repeated across a model family such as Pythia to answer the second question.

OP 6. Fractal trainability of transformers. Does the fractal boundary between convergent and divergent hyperparameters (Sohl-Dickstein, 2024) persist for transformers trained with Adam, and how does its dimension depend on depth? Fine grids over learning rate and warmup length for small transformers are enough for a first estimate.

OP 7. Metastability in generation. Is the length of a coherent stretch of generated text before drift a metastable residence time, exponentially long in some barrier, like the cluster states in depth (Geshkovski et al., 2024)? Survival curves of coherence, judged by an external model, as a function of temperature would test this.

OP 8. Koopman modes of a forward pass. Can a finite-dimensional linear operator fitted to residual-stream trajectories predict the effect of an intervention several layers ahead, and do its eigenfunctions align with known features? Dynamic mode decomposition on residual streams, compared with the tuned lens (Belrose et al., 2023), is a direct test.

OP 9. Targeted energy transfer (conjecture). In coupled mechanical oscillators, a nonlinear energy sink absorbs vibration energy from a primary system in one direction and irreversibly (Vakakis et al., 2009; Farid & Gendelman, 2016). We conjecture that a similar one-way transfer of signal energy exists between spectral modes of the residual stream, with massive activations and attention sinks (Sun et al., 2024; Xiao et al., 2023) in the role of the sink. Tracking the energy carried by each singular mode of the layer Jacobians across depth and training, and testing whether transfer into the sink directions is reversible, would confirm or refute this.

OP 10. Escape under internal forcing (conjecture). At the edge of stability the iterate oscillates along the top Hessian direction, and the other directions see this oscillation as periodic forcing. In mechanics, a periodically forced particle escapes from a potential well when the forcing amplitude crosses a threshold that depends on the forcing frequency (Farid & Gendelman, 2019; Farid, 2022). We conjecture that some loss spikes are escapes of this kind, so that their onset can be predicted from the oscillation amplitude and the local shape of the basin. Gradient descent on a quartic well coupled to an oscillating direction is the simplest test, followed by the same diagnostic on small transformers.

OP 11. Early warning signals. Near a bifurcation, recovery from perturbations slows down and fluctuations grow. Do grokking, induction-head formation and loss spikes show rising autocorrelation and

variance of the loss or of the gradient norm before they happen? Training runs in which the transition step is known, such as those of Power et al. (2022) and Olsson et al. (2022), allow a direct check.

OP 12. Control of clock 3. Can a controller that adjusts the temperature or the hidden state from the stay probabilities equation 6 keep generation out of loops at a lower cost in quality than repetition penalties? Mirostat (Basu et al., 2020) and truncation sampling (Hewitt et al., 2022) at matched quality are the natural baselines.

9 Conclusion

A transformer language model is iterated in three ways: over layers, over optimizer steps and over generated tokens. Each iteration has been studied with tools from dynamical systems, mostly by different communities and under different names. We gave the three iterations one formal frame, reviewed the results on each, and summarized them in a dictionary that records which concepts have been proved, measured or only conjectured on which clock. In three small experiments we found that training makes each layer of a Pythia model more expansive while the coherence of the whole forward pass falls, that gradient descent on a small transformer stays at the edge of stability across two decades of step size, and that repetition loops in GPT-2 and Pythia behave as wells whose depth grows while the model sits in them.

Many cells of the dictionary are empty, and several of the open problems sit where two clocks meet, such as the effect of training on the depth dynamics, the feedback of generated text into training, and the prediction of training instabilities with the tools used for escape problems in mechanics. We hope the shared vocabulary makes it easier for researchers in machine learning and in nonlinear dynamics to work on these questions together.

Code and data

Code to reproduce every figure and number, the raw results, and a maintained list of the papers in this review are available at `https://github.com/maorfarid1/three-clocks`. All experiments run on a CPU.

Broader impact statement

This is a review with small-scale experiments on public models, and we do not foresee direct negative consequences. A better account of repetition loops and of training instabilities could reduce wasted compute in pretraining and make generation more reliable.

Use of AI tools

An AI assistant (Claude, Anthropic) was used to search and verify the literature, draft and edit text, and write and run experiment code under the author's direction. All references were checked against arXiv or Crossref records, all numbers come from the released code, and the author is responsible for every claim in the paper.

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A Experimental details

All experiments run on a CPU with two cores and 7 GB of memory, using PyTorch and Hugging Face Transformers with the public checkpoints of GPT-2 small (Radford et al., 2019) and Pythia (Biderman et al., 2023). The repository contains one script per experiment, the exact command lines, the raw JSON outputs and the plotting scripts. The 40 prompts used in Experiments 1 and 3 are short neutral sentence openings written for this study (for example, "The weather today is cold and cloudy, with a chance of"), and the full list is in the repository.

A.1 Experiment 1: finite-time Lyapunov exponents

For each prompt we take the first T tokens (prompts shorter than T are repeated) and run the model once to record the residual stream $x^0, \dots, x^L$ before the final layer normalization. We then call each block of the Hugging Face implementation directly as a function $F_\ell : \mathbb{R}^{T\times d} \to \mathbb{R}^{T\times d}$, with rotary embeddings and an explicit causal mask, and check that chaining these calls reproduces the model's hidden states (the maximum relative deviation was zero in float32 for every checkpoint). The largest singular value of the Jacobian $DF_\ell(x^\ell)$ is estimated by power iteration on $DF_\ell^\top DF_\ell$, using forward-mode (JVP) and reverse-mode (VJP) products from `torch.func`. The same procedure applied to the composition $F_{L-1} \circ \cdots \circ F_0$

gives the depth-averaged exponent $\frac{1}{L}\log\sigma_{\max}(DF_{0\to L})$. We use eight prompts, contexts of 16 tokens and 10 power iterations per estimate, for eleven Pythia-70M checkpoints (steps 0, 128, 512, 1k, 2k, 4k, 8k, 16k, 32k, 64k and 143k) and six Pythia-160M checkpoints (steps 0, 512, 2k, 8k, 32k and 143k).

A.2 Experiment 2: bifurcation diagram

Data. A first-order Markov chain over $V = 12$ symbols, with transition rows drawn from a Dirichlet distribution with concentration 0.3 (seed 0), generates $N = 64$ sequences of length 13, and the model predicts the next symbol at each of the 12 positions. *Model.* One pre-LN transformer block with causal attention (two heads, width 32, MLP width 64), learned positional embeddings and a linear readout, in float64. Attention is written out explicitly so that Hessian-vector products work on a CPU. *Training.* Full-batch gradient descent on the mean cross-entropy, from the same initialization for every step size within a seed. Seeds 0, 1 and 2 change both the Markov chain and the initialization. Each run lasts $\max(3000, 600/\eta)$ steps, capped at 30,000, so that all runs cover a comparable amount of gradient-flow time. A run is marked as diverged if the loss exceeds 50 or becomes non-finite. *Measurements.* The last 400 losses of each run, and the top Hessian eigenvalue at the final iterate from 40 steps of power iteration on Hessian-vector products. The step sizes are 56 values spaced geometrically between 0.02 and 4.

A.3 Experiment 3: escape from repetition loops

Loop detection. From each prompt we decode greedily for 320 tokens and store the full logit vector at each step. A loop of period $k \le 64$ is detected when the last $3k$ tokens are k-periodic, and its onset is the earliest index from which the sequence stays k-periodic. *Stay probabilities.* Along the loop the context is deterministic, so the probability equation 6 of producing the loop token at temperature T follows from the stored logits, for 30 temperatures spaced geometrically between 0.15 and 2. From these we compute, for each cycle c, the barrier Δ_c (smallest gap between the loop token's logit and any other logit during the cycle) and the escape probability $r_c(T) = 1 - \prod_{t\in c} p_t(T)$. *Arrhenius fit.* For the first cycle after two completed cycles of burn-in, we fit $\log r$ against $1/T$ by least squares over the temperatures where $10^{-9} < r < 10^{-2}$, and compare the negative slope with $\Delta_{\min}$ of that cycle. *Monte Carlo check.* For the first four loops we sample 128 continuations at $T \in \{0.6, 1.0\}$ (nearest grid temperature) from the context after burn-in, record the first step whose token differs from the loop, and compare the fraction still in the loop after 1, 2 and 4 cycles with the exact survival probability. *Free sampling.* For each of seven temperatures we sample 200 tokens from every prompt with plain temperature sampling (no top-k or top-p) and record whether the output ends in a loop with at least three repetitions.